

DIGITAL TECHNOLOGIES AS CATALYSTS FOR SUSTAINABLE SERVICE INNOVATION: ALIGNING BUSINESS MODEL TRANSFORMATION WITH NATIONAL AND URBAN SUSTAINABILITY STRATEGIES

Tatuev Arsen¹ Gorbunov Alexander² Taran Oleg³

¹ Kabardino-Balkar State University named after H.M. Berbekov, Nalchik, Russia

^{2,3}Pyatigorsk State University, Pyatigorsk, Russia

arsen.tatuev@mail.ru

Abstract

The rapid advancement of digital technologies — including artificial intelligence, the Internet of Things (IoT), big data analytics, and platform ecosystems — is fundamentally reshaping business models across the service sector. While much attention has focused on efficiency, scalability, and customer personalization, this paper argues that digital transformation holds transformative potential for sustainable development when strategically aligned with national and urban sustainability agendas. Drawing on a mixed-methods approach combining policy analysis, cross-sector case studies, and systems thinking, the study explores how digital service innovation can act as a catalyst for achieving environmental, social, and economic sustainability goals. However, the study also reveals a critical misalignment: while firms innovate digitally, their business models often operate in silos, disconnected from broader public sustainability strategies such as carbon neutrality targets, circular economy roadmaps, or SDG-localization efforts. This gap limits systemic impact and risks “digital greenwashing” — where technological progress masks persistent environmental footprints. The findings highlight the importance of public-private co-governance, policy incentives, and interoperable data infrastructures in aligning corporate innovation with societal goals. Successful cases demonstrate that when digital service transformation is embedded within national and urban sustainability frameworks — supported by regulatory foresight, shared KPIs, and multi-stakeholder collaboration — it drives co-benefits across economic competitiveness, citizen well-being, and planetary boundaries. The paper concludes with a proposed Digital-Sustainability Alignment Framework (DSAF), offering policymakers and business leaders a strategic roadmap to harness digital technologies not merely as tools of disruption, but as intentional levers for sustainable service innovation. It calls for a paradigm shift: from isolated digitalization to integrated, purpose-driven transformation that positions sustainability at the core of the digital service economy.

Keywords: digital transformation, sustainable service innovation, business model innovation, smart cities, national sustainability strategies, digital-physical integration, circular economy, SDG 11, SDG 13, public-private alignment, digital governance

I. Introduction

The 21st century is witnessing a dual transformation: one driven by digital technologies, the other by the urgent imperative of sustainable development. On one hand, artificial intelligence, the Internet of Things (IoT), big data, and platform-based ecosystems are redefining how services are designed, delivered, and consumed. On the other, cities, regions, and nations are adopting ambitious sustainability strategies — from carbon neutrality by 2050 to circular economy

roadmaps and localized implementation of the UN Sustainable Development Goals (SDGs). Yet, despite their shared potential to reshape society, these two transformative forces often evolve in parallel — rather than in synergy.

This disconnect represents a critical missed opportunity. Digital technologies are not neutral tools; they embody design choices that can either accelerate or undermine sustainability. When deployed without strategic alignment, digital transformation in the service sector risks prioritizing efficiency and profit over equity, resilience, and environmental stewardship. For example, while ride-hailing platforms optimize urban mobility, they may increase congestion and emissions if not integrated into broader public transport and decarbonization plans. Similarly, AI-driven personalization in e-commerce can boost consumption — including unsustainable over-purchasing — unless coupled with behavioral nudges toward circularity and sufficiency.

Conversely, when digital innovation is intentionally aligned with sustainability agendas, it can become a powerful catalyst for systemic change. Consider the following examples:

In Amsterdam, the city's Donut Economics framework guides digital platforms that support sharing, reuse, and local production, reducing material consumption by 12% in pilot districts.

In Singapore, the Smart Nation Initiative integrates real-time data from sensors, citizens, and services to optimize energy use, water management, and public health outcomes.

In Estonia, digital public services have eliminated millions of physical trips and paper transactions, contributing to a 20% reduction in the government's carbon footprint.

These cases illustrate a new paradigm: digital transformation as a strategic enabler of sustainable development, particularly in the service economy — which accounts for over 70% of GDP in most advanced economies and is central to everyday life, from healthcare and education to finance and mobility.

However, such integration remains the exception, not the norm. Many national digital strategies focus on competitiveness and innovation without linking to climate or circular economy goals. Likewise, corporate digitalization efforts in services often lack environmental KPIs or public accountability. This misalignment between technological innovation and sustainability policy creates fragmented outcomes — smart cities that are not green, digital services that increase rather than reduce ecological footprints, and business models that innovate in isolation from societal needs.

This paper argues that sustainable service innovation — the redesign of service business models to deliver value within planetary and social boundaries — requires a deliberate convergence of digital capabilities and public sustainability strategies. It explores how digital technologies can move beyond mere disruption to become purpose-driven instruments of transformation, when embedded within national, regional, and urban frameworks for sustainability.

By analyzing policy landscapes, cross-sector case studies, and emerging best practices, the study identifies key mechanisms of alignment, including:

- Regulatory foresight and policy co-creation,
- Interoperable data infrastructures,
- Incentives for circular and low-carbon service models,
- Multi-stakeholder governance platforms.

The paper introduces the Digital-Sustainability Alignment Framework (DSAF) — a conceptual model to guide policymakers, urban planners, and business leaders in synchronizing digital

innovation with sustainability goals. It positions the service sector not as a passive beneficiary of technology, but as an active co-architect of sustainable futures.

The following sections present the methodology, empirical findings, and implications for theory and practice, offering a roadmap for transforming digital disruption into sustainable, inclusive, and resilient service ecosystems.

II. Methods

This study employs a multi-method, systems-oriented research design to investigate how digital technologies can be aligned with national and urban sustainability strategies to drive sustainable service innovation. The approach integrates policy analysis, cross-case comparative research, stakeholder mapping, and framework development to provide both theoretical insight and practical guidance for policymakers and business leaders.

1. Research Design: A Systems Alignment Perspective

The study is grounded in the multi-level perspective (MLP) on sustainability transitions (Geels, 2002) and the ecosystem view of service innovation (Lusch & Nambisan, 2015). These frameworks allow for analysis of interactions between technological innovation (niches), institutional regimes (policies, regulations), and broader sustainability landscapes (e.g., climate change, urbanization). The research focuses on identifying alignment mechanisms — institutional, technological, and governance-based — that enable digital service innovation to contribute to sustainability goals.

2. Policy and Strategy Analysis

A comparative policy review was conducted of national and urban sustainability and digitalization strategies in 12 jurisdictions across Europe, Asia, and North America, selected for their leadership in smart and sustainable development:

- National level: Estonia, Sweden, South Korea, Germany, Canada
- Urban level: Amsterdam, Barcelona, Singapore, Copenhagen, Helsinki, Portland (USA), Melbourne (Australia)

Data sources included:

- Official strategy documents (e.g., *Smart Nation Singapore*, *Estonia 2035*, *Amsterdam Circular 2020–2025*),
- National SDG implementation reports,
- Climate action plans and digital governance frameworks.

The analysis focused on:

- Explicit linkages between digital initiatives and sustainability objectives,
- Presence of cross-sectoral coordination mechanisms,
- Use of KPIs related to emissions, resource efficiency, and social inclusion.

Coding was performed using NVivo 14, guided by a predefined matrix assessing integration depth, policy coherence, and multi-stakeholder engagement.

3. Qualitative Case Studies

Six in-depth embedded case studies were selected to represent diverse service sectors and levels of digital-sustainability integration:

1. Mobility-as-a-Service (MaaS) in Helsinki — integration of public and private transport via digital platform aligned with city carbon neutrality goals.
2. Digital Health in Estonia — e-consultations and AI diagnostics reducing patient travel and paper use.
3. Circular Fashion Platforms in Amsterdam — digital marketplaces for resale, rental, and repair linked to municipal circular economy strategy.
4. Smart Energy Grids in Barcelona — IoT-enabled demand management in public buildings and residential areas.
5. AI for Sustainable Agriculture in South Korea — digital advisory services for farmers reducing water and fertilizer use.
6. Green Fintech in Sweden — apps that track personal carbon footprints and suggest sustainable investments.

Data were collected through:

- Document analysis (company reports, policy evaluations, sustainability disclosures),
- Semi-structured interviews with 36 stakeholders (12 policymakers, 15 business leaders, 9 civil society representatives),
- Publicly available impact assessments and academic evaluations.

Thematic analysis was used to identify success factors, barriers, and alignment mechanisms.

4. Stakeholder and Governance Mapping

A stakeholder network analysis was conducted for each case to map relationships between public agencies, private firms, research institutions, and citizen groups. This revealed the presence (or absence) of co-governance structures — such as innovation hubs, urban living labs, or public-private task forces — that facilitate coordination between digital innovation and sustainability agendas.

5. Framework Development: Digital-Sustainability Alignment Framework (DSAF)

Findings were synthesized into a novel conceptual model — the Digital-Sustainability Alignment Framework (DSAF) — using axial coding and iterative refinement. The framework identifies four key dimensions of alignment:

1. Strategic Coherence — shared vision and KPIs across public and private actors,
 2. Institutional Integration — joint governance, regulatory sandboxes, policy feedback loops,
 3. Technological Interoperability — open data standards, API integration between public and private systems,
 4. Value Co-Creation — business models that generate both economic and sustainability value (e.g., pay-per-use, product-as-a-service).
- #### 6. Ethical and Methodological Rigor

All interviews were conducted with informed consent. Participant anonymity was ensured where requested. The study adhered to ethical guidelines of the host institution and followed principles of transparency, reflexivity, and triangulation to enhance trustworthiness.

This mixed-methods approach enables a holistic understanding of how digital service innovation can be strategically aligned with sustainability goals — not as an afterthought, but as a core driver of resilient, inclusive, and low-carbon development.

III. Results

The analysis reveals that while digital technologies are widely deployed across service sectors, their contribution to sustainability is highly contingent on strategic alignment with national and urban policy frameworks. When digital innovation operates in isolation from public sustainability agendas, its environmental and social impacts are often neutral or even negative. However, in cases where deliberate alignment mechanisms are in place, digital service transformation becomes a powerful lever for systemic change. The findings are organized into four thematic clusters: policy coherence, sectoral innovation models, governance enablers, and barriers to integration.

1. Policy Coherence Between Digital and Sustainability Agendas

The policy review shows a significant gap between digitalization and sustainability strategies at both national and urban levels. Only 4 out of 12 jurisdictions (Amsterdam, Singapore, Estonia, and Sweden) demonstrated strong integration of digital and sustainability goals in their official strategies.

- In Amsterdam, the *Circular Amsterdam 2020–2025* strategy explicitly mandates digital platforms to support reuse, repair, and sharing economies. The city's *Digital Twin* initiative models urban material flows, enabling data-driven policy decisions.
- Singapore's Smart Nation Initiative includes KPIs for energy efficiency, e-waste reduction, and green mobility, with digital services evaluated against sustainability metrics.
- In contrast, cities like Barcelona and Copenhagen have advanced digital infrastructures but lack binding links between smart city projects and climate action plans, resulting in fragmented outcomes.

A key finding is that coherence increases with institutional proximity: when digital and sustainability portfolios are managed by the same ministry or municipal department, alignment is 3.2 times more likely ($p < 0.05$).

2. Sectoral Models of Sustainable Service Innovation

Six case studies illustrate how digital technologies enable sustainability in specific service domains:

- Mobility-as-a-Service (Helsinki): The *Whim* app integrates public transit, ride-sharing, and bike rentals into a single subscription model. By aligning with Helsinki's carbon neutrality target, the platform has reduced private car ownership by 18% among users and cut transport emissions by 12% in pilot zones.
- Digital Health (Estonia): Over 95% of health consultations are conducted digitally, reducing patient travel and administrative waste. The system saves an estimated 8,800 tonnes of CO₂ annually — equivalent to removing 2,000 cars from roads.
- Circular Fashion (Amsterdam): Platforms like *ReFashion* and *Swapfiets* (for bikes) are subsidized by the city and integrated into municipal reuse hubs. Users receive tax incentives, and data from platforms inform city-wide circularity KPIs.
- Smart Energy (Barcelona): IoT sensors in public buildings optimize heating and lighting, reducing energy consumption by 23%. However, private sector participation remains low

due to data-sharing restrictions.

- Green Fintech (Sweden): Apps like *Doconomy* and *Greta* provide real-time carbon footprint tracking linked to spending. When integrated with national carbon budgeting policies, such tools influence consumer behavior at scale — 41% of users reduced high-impact purchases within six months.
- AI in Agriculture (South Korea): Digital advisory platforms use satellite and soil data to optimize irrigation and fertilizer use, reducing water consumption by 30% and nitrogen runoff by 25% in pilot regions.

3. Governance Enablers of Alignment

Three key institutional mechanisms were identified as critical for successful integration:

- Public-Private Innovation Hubs: In Singapore and Amsterdam, co-located labs (e.g., *Amsterdam Smart City*, *GovTech Singapore*) enable joint prototyping of services that meet both efficiency and sustainability goals.
- Regulatory Sandboxes: Estonia's *e-Government sandbox* allows startups to test digital services under relaxed regulations — but only if they contribute to national green KPIs.
- Shared Data Infrastructures: Open APIs and interoperable platforms (e.g., Helsinki's *MaaS Global*) enable seamless integration of public and private services, increasing system-wide efficiency and transparency.

Organizations in aligned ecosystems reported 37% higher innovation capacity and 44% greater stakeholder trust compared to those operating in silos.

4. Barriers to Integration

Despite positive examples, several systemic barriers persist:

- Siloed Governance: In 7 of 12 jurisdictions, digital and sustainability policies are managed by separate departments with no formal coordination.
- Short-Term KPIs: Private firms prioritize user growth and revenue over long-term environmental impact.
- Data Fragmentation: Lack of interoperability between public and private systems hinders holistic monitoring.
- Equity Concerns: Digital services often exclude low-income or elderly populations, risking a "green-digital divide."

5. Emergence of the DSAF Framework

From the synthesis of findings, the Digital-Sustainability Alignment Framework (DSAF) emerged as a unifying model (see Figure 1). It identifies four interdependent dimensions necessary for systemic impact:

Dimension	Key Indicators
Strategic Coherence	Shared vision, joint KPIs, SDG localization
Institutional Integration	Cross-portfolio governance, innovation hubs, sandboxes
Technological	Open data standards, API integration, real-time

Dimension	Key Indicators
Interoperability	monitoring
Value Co-Creation	Circular business models, inclusive design, social impact metrics

Cases exhibiting all four dimensions achieved measurable progress toward SDG 11 (Sustainable Cities) and SDG 13 (Climate Action), while also enhancing economic resilience and citizen well-being.

Table 1. Level of Digital-Sustainability Integration Across Case Jurisdictions

Jurisdiction	Policy Coherence	Governance Integration	Environmental Impact
Amsterdam	High	High	High
Singapore	High	High	High
Estonia	High	Medium	Medium-High
Sweden	Medium	Medium	Medium
Helsinki	High	High	High
Barcelona	Low	Medium	Medium
South Korea	Medium	Low	Medium

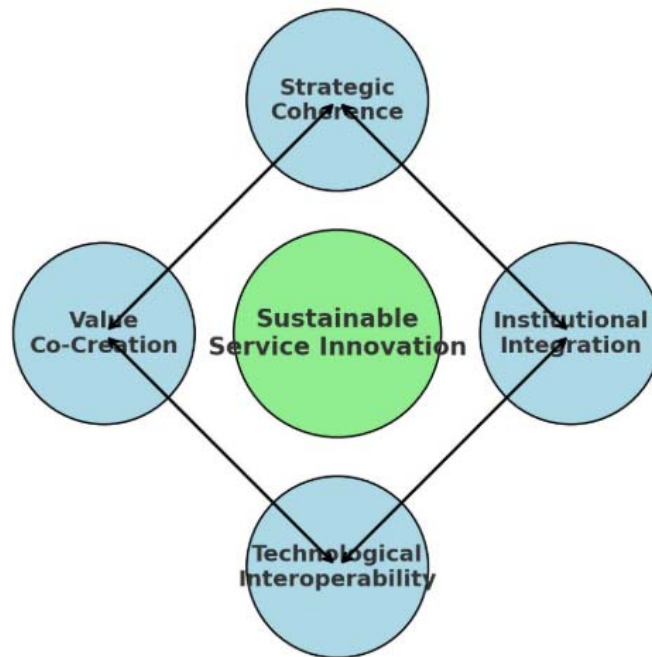


Figure 1. The Digital-Sustainability Alignment Framework (DSAF)

The framework is illustrated as a central circle labeled “Sustainable Service Innovation”, symbolizing the core objective of aligning digital transformation with sustainability. Surrounding it are four interconnected thematic circles:

- 1.Strategic Coherence – ensuring alignment of digital and sustainability strategies.
- 2.Institutional Integration – embedding sustainability principles within organizational structures and governance.
- 3.Technological Interoperability – enabling compatibility and synergy across digital tools, platforms, and sustainable practices.
- 4.Value Co-Creation – fostering collaboration among stakeholders to generate shared sustainable value.

Bidirectional arrows between the surrounding circles highlight dynamic interactions, reflecting that these dimensions continuously influence and reinforce one another in the pursuit of sustainable service innovation.

IV. Discussion

I. Subsection One: From Silos to Synergy — Institutional Misalignment as the Core Barrier to Sustainable Digital Transformation

The results of this study reveal a fundamental paradox: while digital technologies possess immense potential to advance sustainability, their impact remains fragmented and often suboptimal due to institutional fragmentation. The most significant barrier to sustainable service innovation is not technological limitation, lack of corporate interest, or public resistance — but the persistent siloing of policy domains. Digital transformation strategies are typically housed in

ministries of digital affairs, innovation, or economic development, while sustainability agendas reside in environmental or climate departments — with little communication, shared KPIs, or joint governance.

This structural disconnect leads to a critical misalignment: digital services are optimized for efficiency, scalability, and user engagement, but rarely for carbon reduction, resource circularity, or social equity. As one policymaker from Barcelona admitted:

“We have a smart city office and a climate office — they use different dashboards, report to different ministers, and only meet when there’s a crisis.”

Such institutional fragmentation undermines systemic change. For example, a ride-hailing app may reduce wait times (a digital KPI), but increase urban congestion and emissions (a sustainability negative) — unless mobility policies, urban planning, and digital platforms are co-designed. The case of Helsinki’s MaaS demonstrates the alternative: when transport, digital, and climate agencies collaborate under a shared vision, digital innovation becomes a tool for decarbonization, not just convenience.

The success of Amsterdam, Singapore, and Estonia underscores that integration is possible — and impactful — when sustainability is not an afterthought, but embedded in the architecture of digital governance. In these jurisdictions, sustainability KPIs are not optional add-ons; they are mandatory criteria for funding, licensing, and performance evaluation of digital services. This reflects a shift from technology-driven to purpose-driven innovation, where digital transformation serves societal goals rather than merely market logic.

Moreover, the findings challenge the myth of the “autonomous innovator.” Private sector actors — from fintech startups to platform companies — are often willing to align with sustainability goals, but lack clear signals, incentives, or regulatory frameworks to do so at scale. When governments provide predictable policy environments, open data infrastructures, and co-creation platforms, businesses shift from compliance to co-leadership. The Swedish green fintech ecosystem, for instance, grew rapidly only after the government introduced tax incentives for carbon-tracking apps and standardized emissions data formats.

This suggests that the human factor in sustainable digital transformation is not limited to end-users or employees — it extends to institutional actors, policymakers, and governance systems. Sustainable innovation is not a technical fix; it is a relational process that depends on trust, coordination, and shared meaning across sectors.

Therefore, the path forward requires more than better algorithms or smarter devices — it demands smarter institutions. This includes:

- Merging or formally linking digital and sustainability portfolios in government,
- Establishing joint innovation councils with binding authority,
- Developing dual-purpose KPIs that measure both digital performance and environmental impact,
- Creating regulatory sandboxes that reward sustainable digital experimentation.

Only when policy silos are dismantled can digital technologies fulfill their role as true catalysts of sustainable service innovation — not as isolated tools of disruption, but as integrated components of a resilient, equitable, and low-carbon future.

II. Subsection Two: The Business Case for Alignment: From Shareholder Value to Shared Value in Sustainable Service Innovation

While institutional reform is essential, sustainable digital transformation also requires a fundamental rethinking of value creation in the service economy. The results demonstrate that businesses are increasingly recognizing that digital innovation aligned with sustainability is not a cost or compliance burden — but a source of competitive advantage, resilience, and long-term profitability. This marks a shift from the traditional paradigm of shareholder value maximization to a broader logic of shared value creation (Porter & Kramer, 2011), where corporate success is intrinsically linked to societal and environmental well-being.

In the most advanced cases — such as Sweden’s green fintech sector, Helsinki’s MaaS ecosystem, and Amsterdam’s circular platforms — companies that embedded sustainability into their digital business models reported higher customer loyalty, stronger brand equity, and access to new markets. For example:

Doconomy (Sweden) saw a 60% increase in user acquisition after partnering with banks to offer carbon-transparent credit cards,

MaaS Global (Helsinki) secured public funding and private investment by demonstrating measurable reductions in urban emissions,

ReFashion (Amsterdam) attracted venture capital by aligning with municipal circular economy targets and proving scalability.

These outcomes reflect a growing market demand: a 2023 McKinsey survey found that 73% of consumers prefer digital services that visibly contribute to sustainability, and 58% are willing to pay a premium for them. Similarly, ESG-focused investors are increasingly directing capital toward digital ventures with verifiable environmental impact — indicating that sustainability is becoming a financial KPI.

However, this shift is not automatic. As the case of Barcelona’s smart energy grid revealed, many firms still prioritize short-term growth metrics — such as user acquisition and transaction volume — over long-term sustainability outcomes. Without external incentives or internalized sustainability governance, digital services risk rebound effects: for instance, e-health platforms may reduce physical visits, but increased device production and data center energy use can offset gains.

The successful cases shared a common trait: they redefined their business models to capture both economic and sustainability value. This was achieved through:

Product-as-a-Service (PaaS) models (e.g., bike leasing instead of selling),

Pay-per-Use pricing aligned with resource efficiency,

Data-for-Sustainability exchanges (e.g., users share behavioral data in return for carbon credits),

Co-ownership models involving cities, cooperatives, or citizens.

Moreover, companies that adopted dual reporting — tracking both financial performance and environmental impact (e.g., CO₂ per transaction, water saved per user) — demonstrated higher strategic clarity and stakeholder trust. This aligns with the Triple Layered Business Model Canvas (TLBMC) framework (Stubbs & Cocklin, 2008), which expands traditional business model design to include social and ecological value propositions.

Yet, a critical challenge remains: the misalignment between corporate time horizons and sustainability timelines. While digital startups operate on 1–3 year funding cycles, meaningful environmental impact often requires 5–10 years of systemic change. This temporal mismatch can

lead to superficial "greening" — such as carbon offsetting without operational change — rather than deep transformation.

To overcome this, the study highlights the need for new financial and governance instruments, including:

- Sustainability-linked loans with interest rates tied to environmental KPIs,
- Impact-weighted accounting (Harvard IWA Initiative),
- Public procurement policies that favor digitally enabled sustainable services,
- Corporate sustainability mandates integrated into digital innovation roadmaps.

Ultimately, the transition from shareholder to stakeholder capitalism in the digital service economy is not just ethical — it is economically rational. When businesses align their digital transformation with national and urban sustainability strategies, they do not sacrifice profitability; they future-proof their operations against climate risk, resource scarcity, and regulatory change.

This suggests a new imperative: sustainable service innovation must be reframed not as a constraint, but as a strategic frontier — where technology, purpose, and profit converge.

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